Executive Summary: Colombian Oil Output under Proved Reserves and Expected Royalties

Final Integrative Project · Rafael Fajardo Moreno, Sonia Catalina Martinez Herrera and Laura Marcela Garnica Tapia · October 2026 · Restricted data, academic use only (EAN University, Machine Learning Specialization, 2026)

Decision supported. Colombia's Ministry of Mines and Energy, the National Hydrocarbons Agency and the National Planning Department budget oil royalties from the operators' Proved Reserves (1P) forecast in the 2024 Reserves and Resources Report (IRR 2024). This project answers how annual and daily oil output under the 1P profile is expected to behave, and how much royalty revenue to expect each year at WTI or Brent prices.

What we did. Following CRISP-DM, we used the official field-level file: 2023 to mid-2026 production plus proved, probable and possible reserves through 2060 for 320 fields. We trained decision trees, linear regression, a neural network and tree ensembles to predict each field's yearly change, validated them by field groups and over time, and tested once on 2026.

Results. The selected random forest cut field-level error to 8.4%, against 25.4% when the 1P figure is used as the forecast, with a national bias of -3.1% (a slight underestimate). Simply repeating last year scored 8.1% on 2026, so the model's added value lies in tracking the 1P decline in later years. Output falls from 746 thousand barrels per day in 2025 to 724 thousand in 2026, 535 thousand in 2027 and 328 thousand in 2030. Royalties under Law 756 of 2002 total about 27.9 million barrels in 2026, worth USD 2.37 billion at WTI or USD 2.60 billion at Brent, and USD 1.62 billion (WTI) in 2027.

Recommendation. Use the model for 12 to 24 month royalty planning and keep 1P as a ceiling. Read the post-2027 path as a conservative floor, since it is capped at proved reserves. Review the fields flagged as at risk of producing below 1P with operators; the alert caught 94% of 2026 shortfalls.

Risks and follow-up. Short history, field mergers, security disruptions, price assumptions beyond 2027, and scope: only the IRR 2024 reserves development forecast, without 2P/3P migration or contingent resources. Retrain yearly, rerun the blind test and revise the model if national bias exceeds 5%. A local app lets decision makers query any year, region, operator or contract type.